

COASTAL DYNAMICS

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- **Outline:**

1. Introduction
2. General equations
3. Surface Waves
 - 3.1. Surface gravity waves
 - 3.2. Inertia-gravity Waves
 - 3.3. Coastal waves
4. Tides
5. Internal Waves
6. Geostrophy and thermal wind
7. Bottom and surface boundary layers
8. Coastal circulation and responses to meteorological forcing
9. Coastal Fronts and river plumes

Presentations and material
will be available at :

jgula.fr/Coastal/

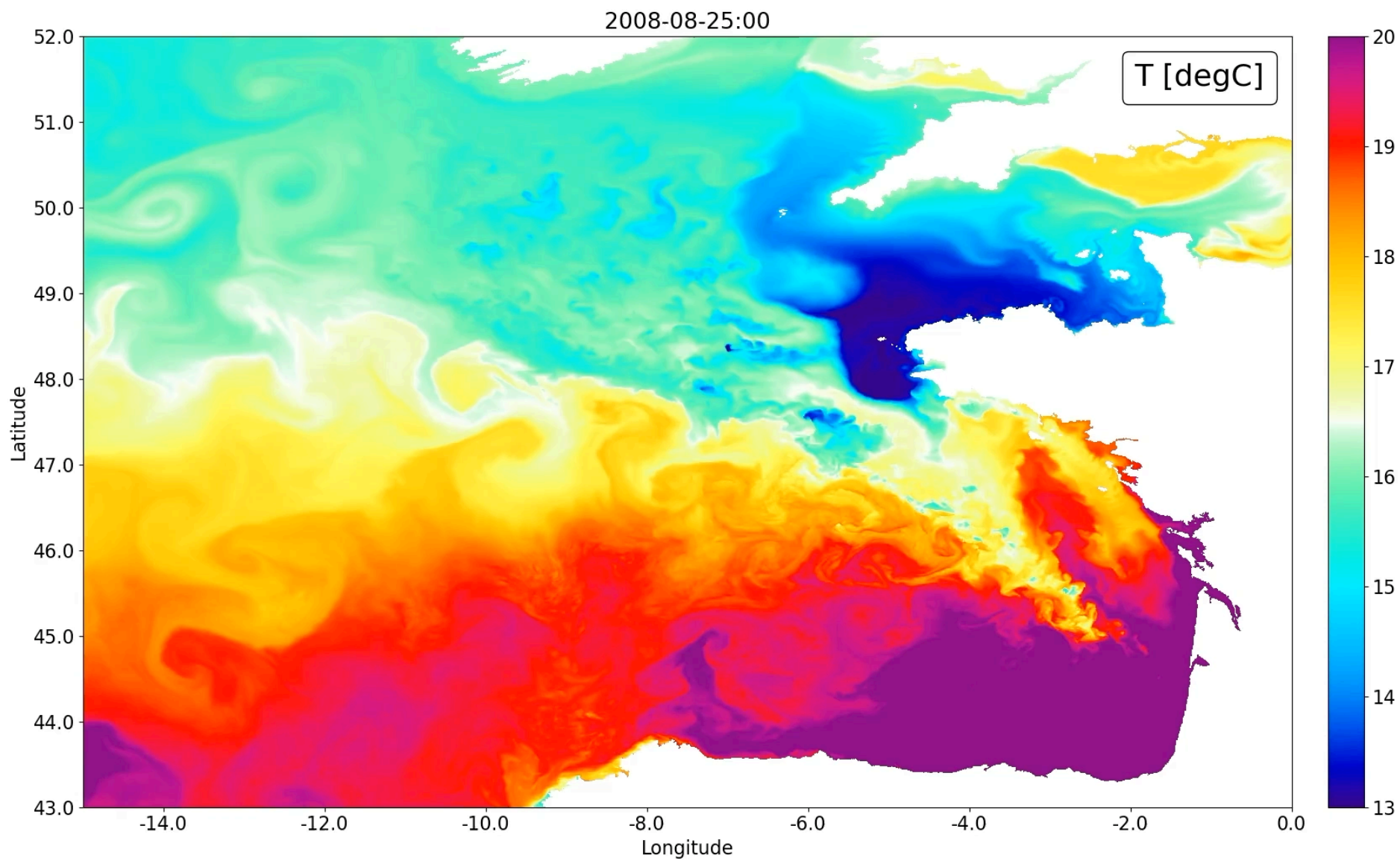
- **Definition**

Region of the ocean where water mass hydrological gradients are strong

- **Causes**

These salinity and temperature gradients are the mark of simultaneous changes of the physical processes that drives the water column structure.

Coastal Fronts



Why these frontal area are of great interests :

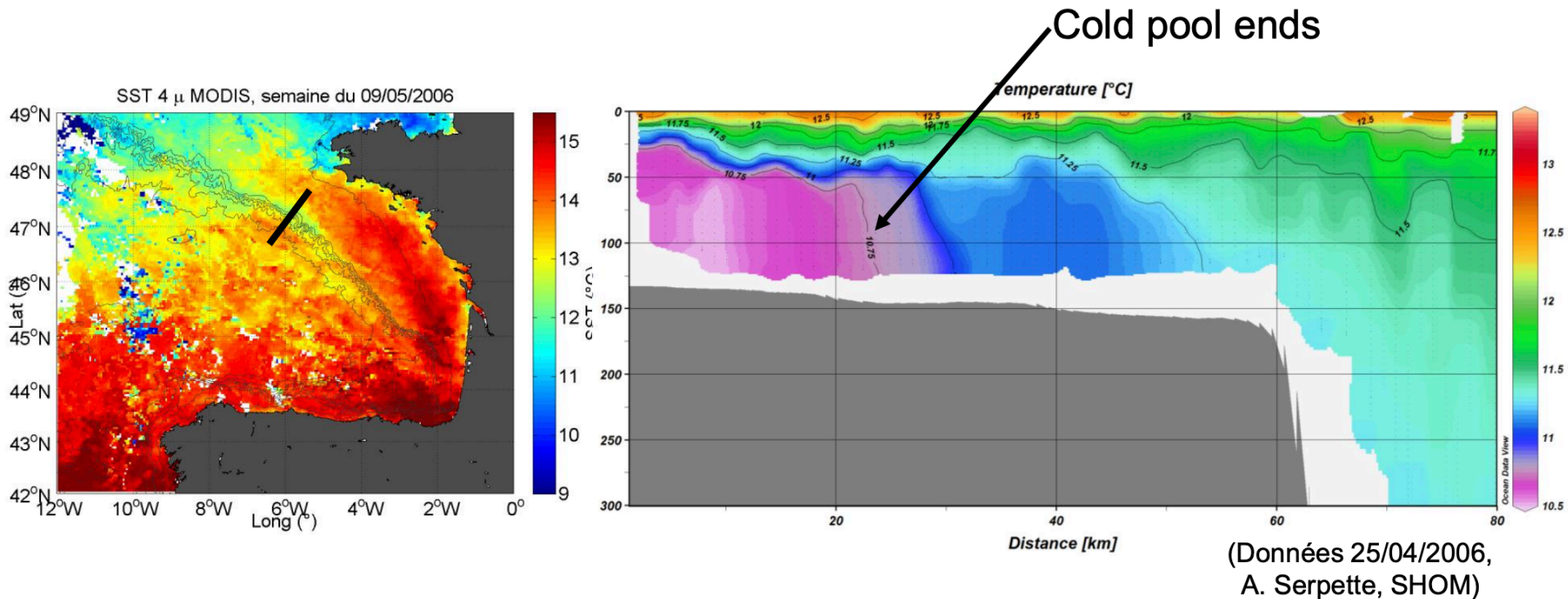
- Associated to rapid changes of biogeochemical equilibrium.
- Places where the primary production is enhanced (due to local mixing and thus fertilization) and consequently secondary production (zooplankton), fishes stocks, birds.... Giving rather more productive ecosystems.

Different types of fronts on the shelf :

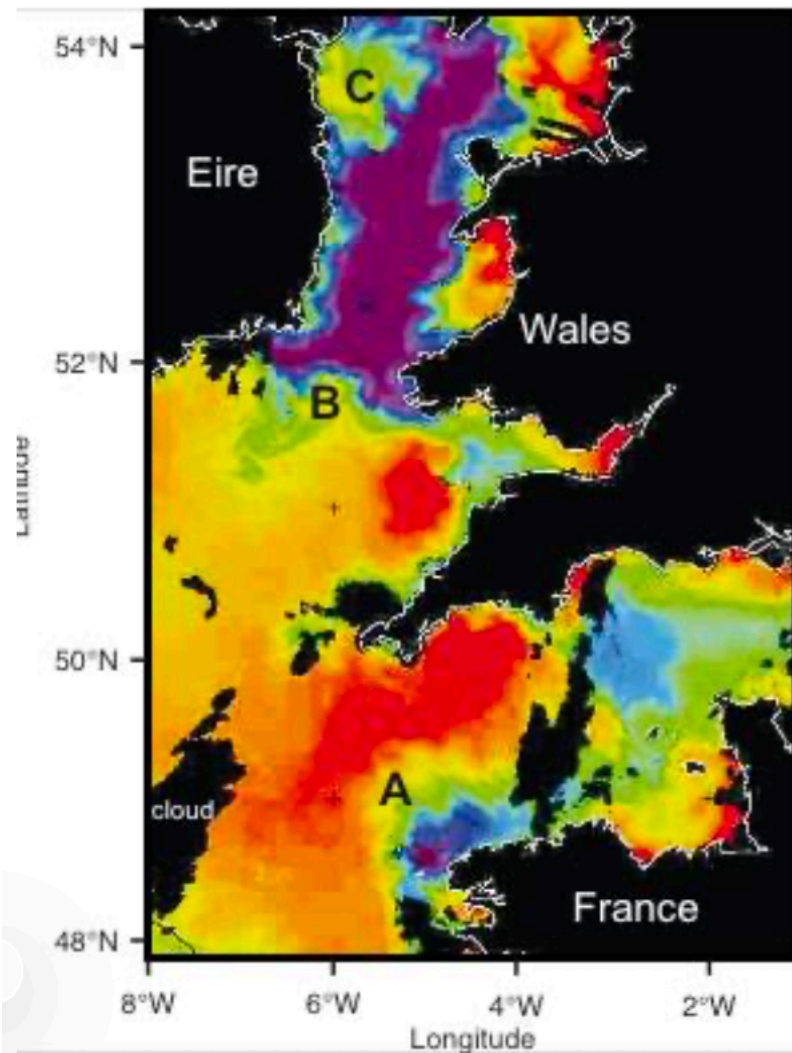
- Induced by the wind (upwelling)
- Different water masses at the shelf break-> **shelf break fronts**
- Induced by buoyancy flux (heat) combined to tidal mixing -> **thermal/tidal fronts**
- influenced by continental runoffs with fresh (and rich) waters -> **river plumes**

Shelf break fronts

The interaction of the tidal wave with the shelf break give rise to an intensified mixed area where the stratification hardly sets on (due to this permanent mixing).

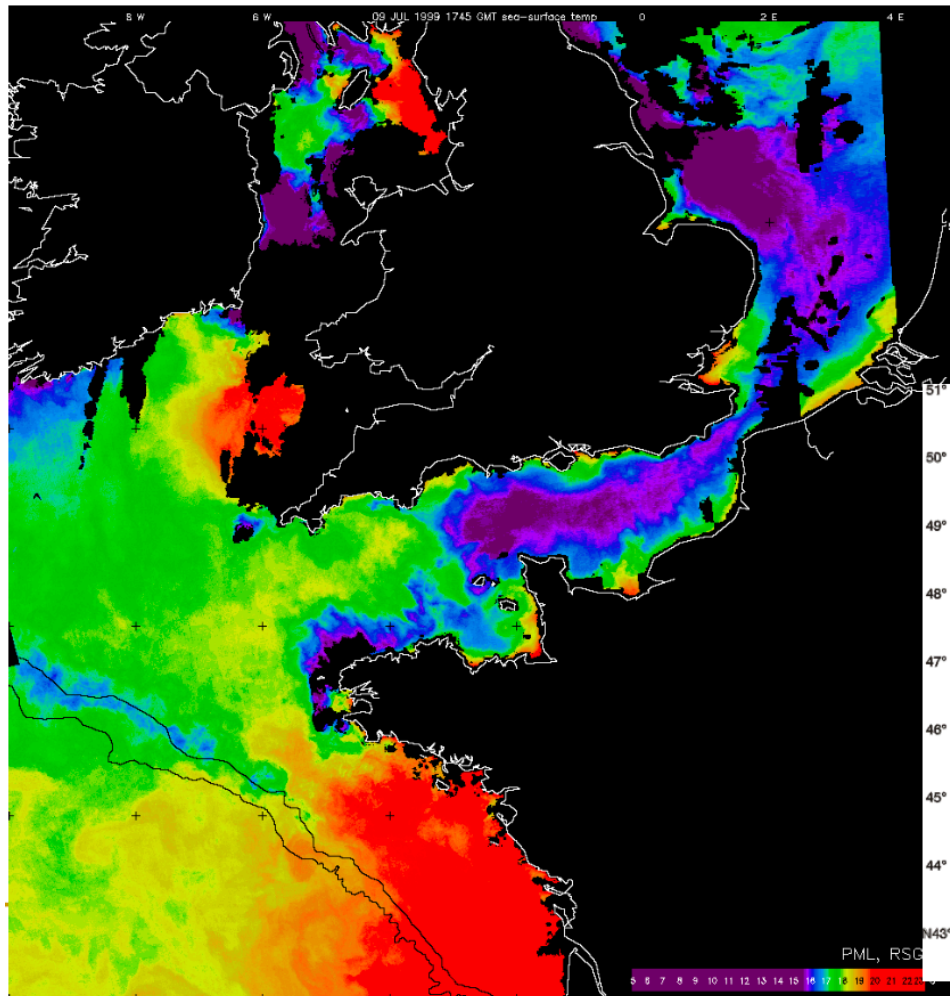


thermal/tidal fronts

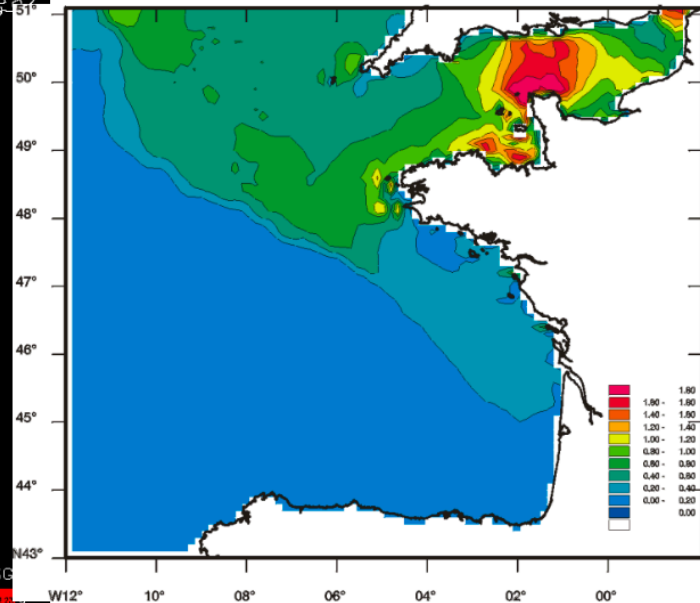


thermal/tidal fronts

Summer scene . Picture of july the 9th 1999



Maximum (over a cycle) tidal current (for a mean tide)

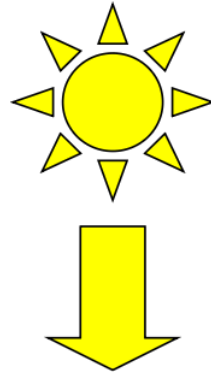


Seasonal stratification



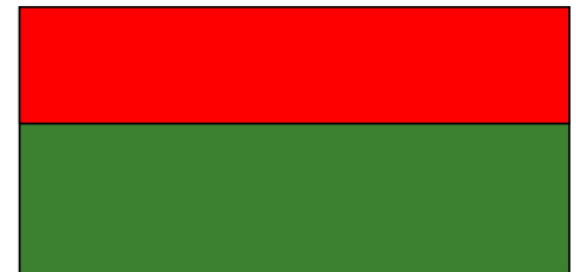
End of winter time:

Cold water is well mixed over the shelf



Spring:

The sun heat the surface layer, which mixes less and less with the bottom.

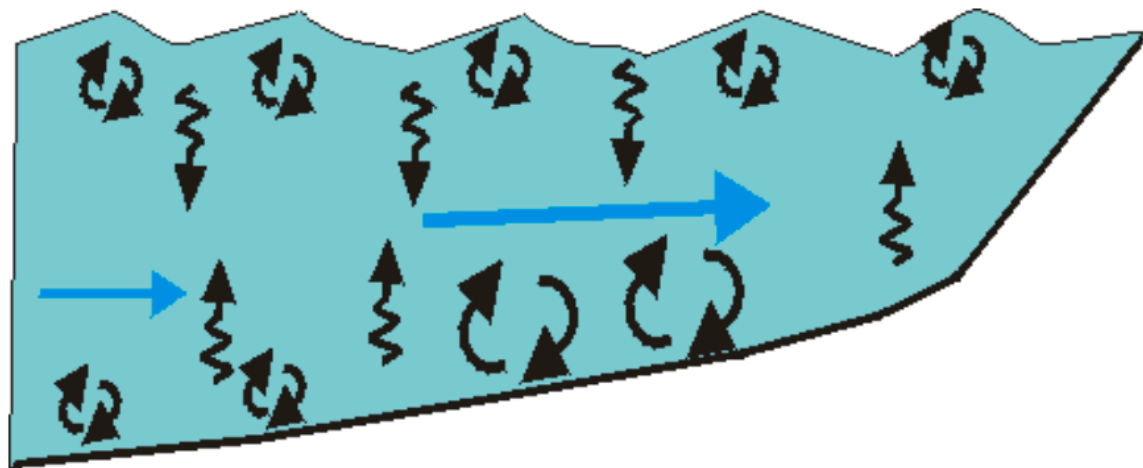


Summer :

At the surface, a warm (~18-20°C) layer (30 m thick).
At the bottom a cold pool (11.5°C).

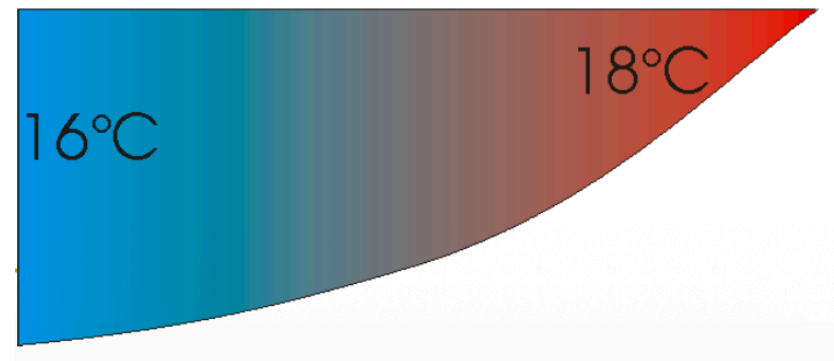
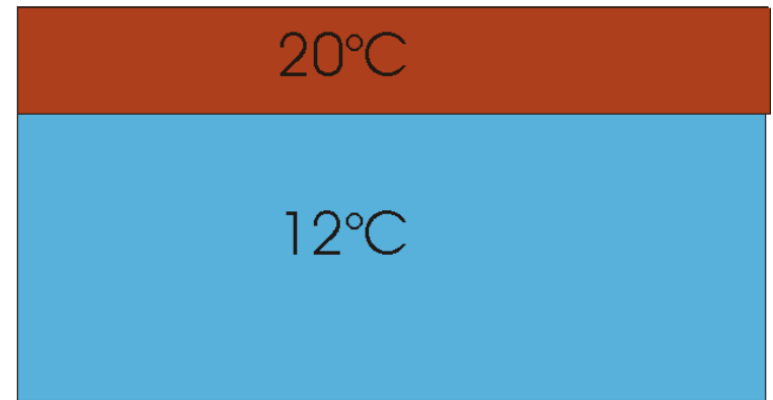
The mixing

- The vertical mixing of the water mass is due to turbulence.
- The turbulence is generated at the surface (wind shear, wave breaking) and at the bottom (velocity shear).

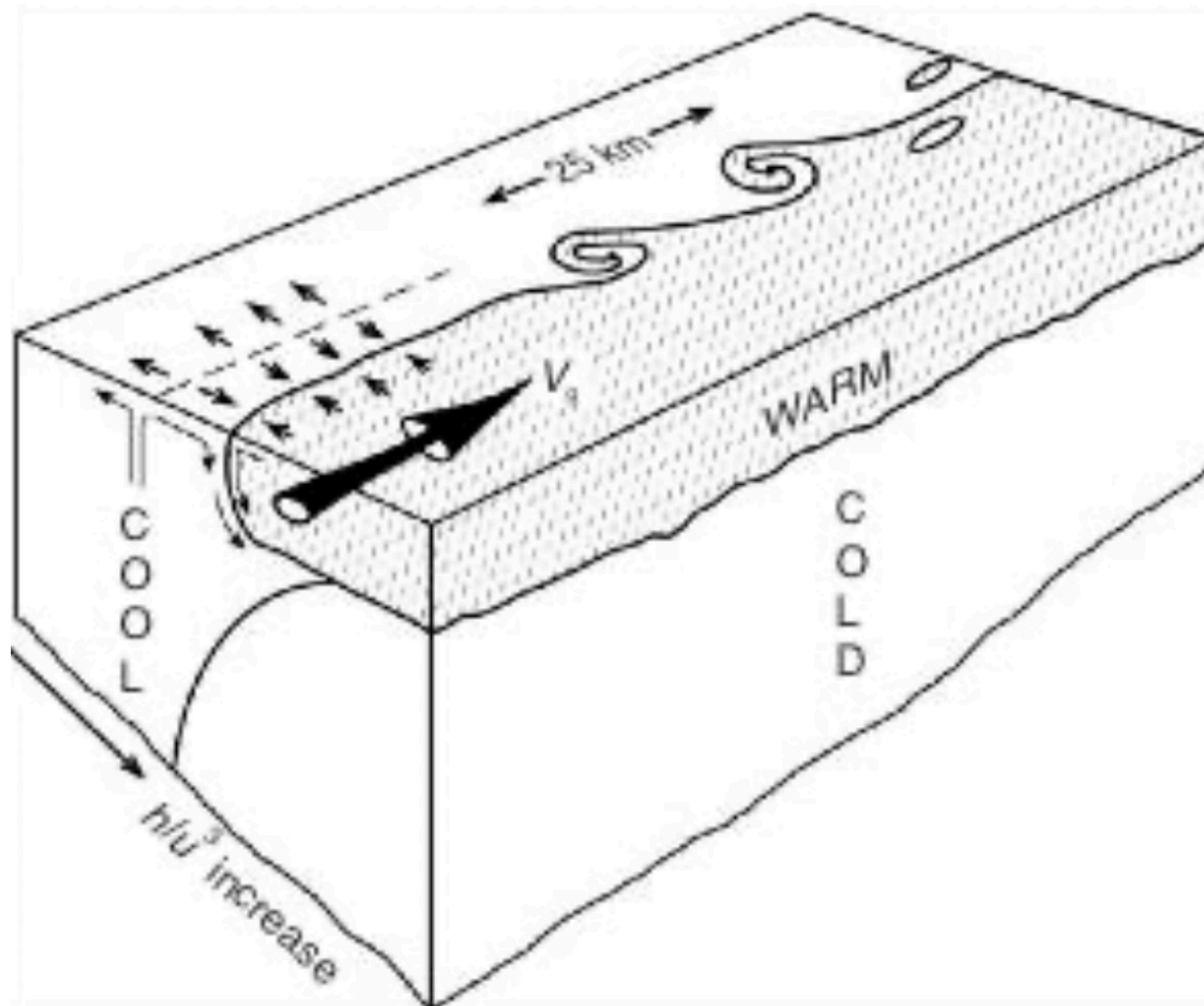


Effect of mixing on temperature

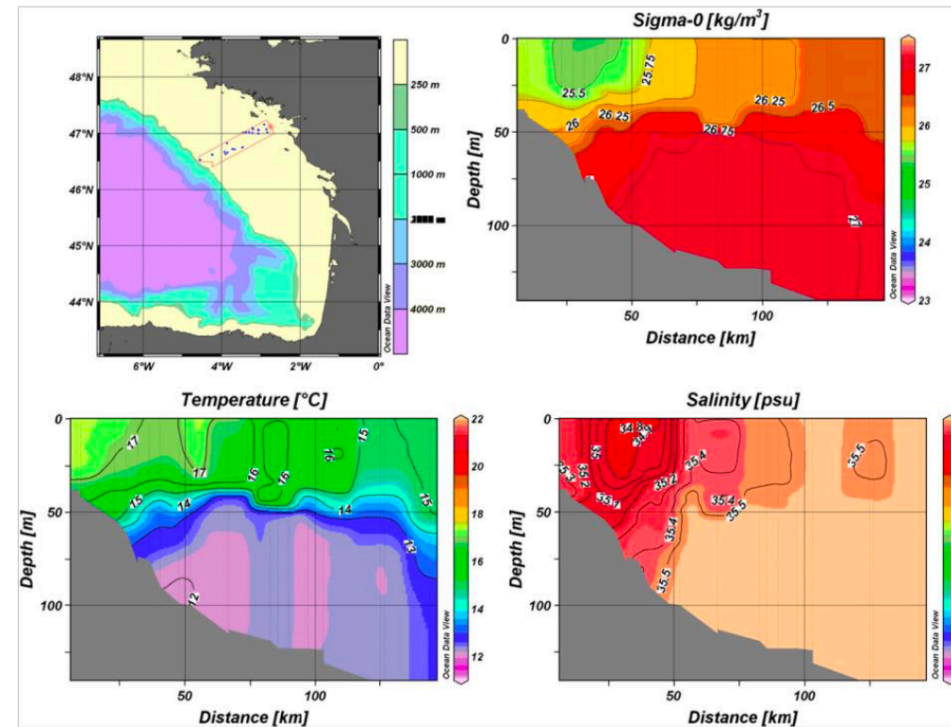
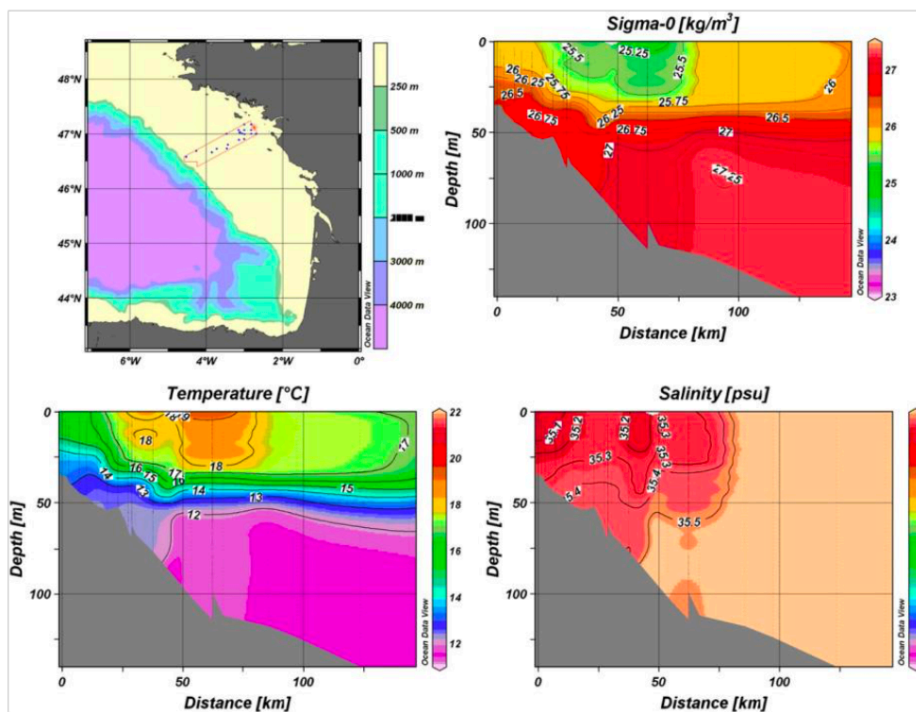
- The mixing is not strong enough to mix up the whole water column => stratification
- The mixing (due to currents and wind) is strong enough to mix the whole water column which thus becomes homogenous. The temperature distribution is tightly linked to the water column height.



Coastal fronts (description)



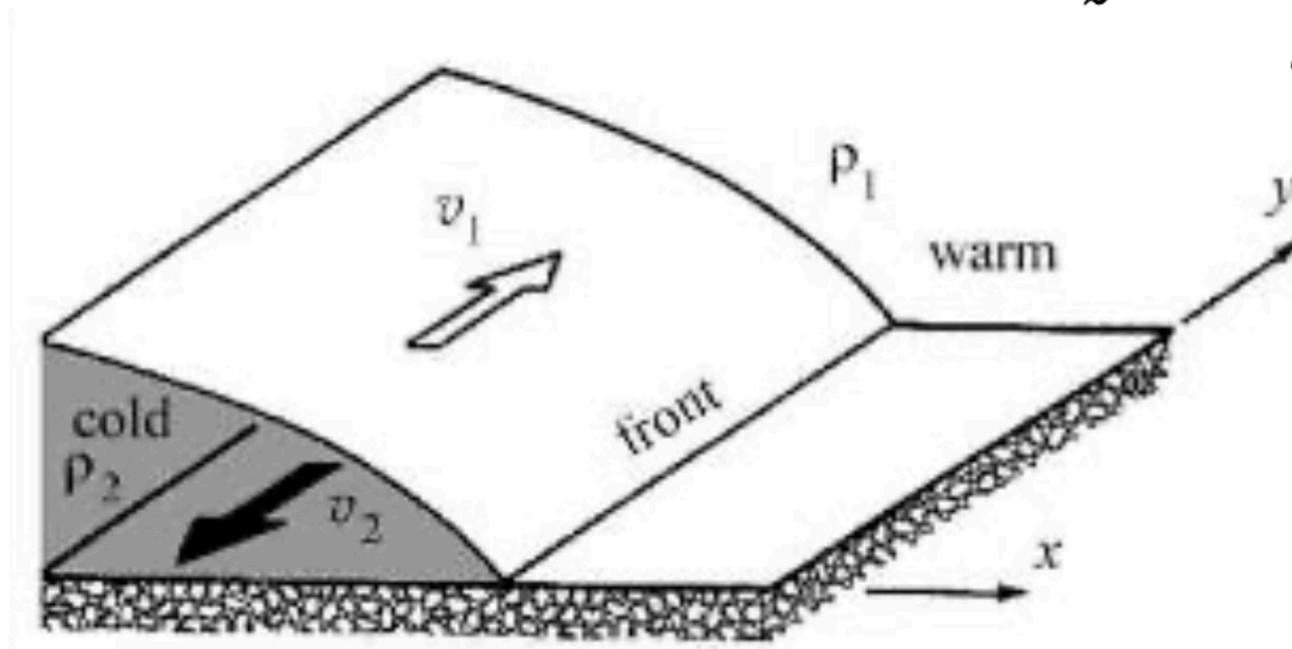
Spatial distribution of the water mass



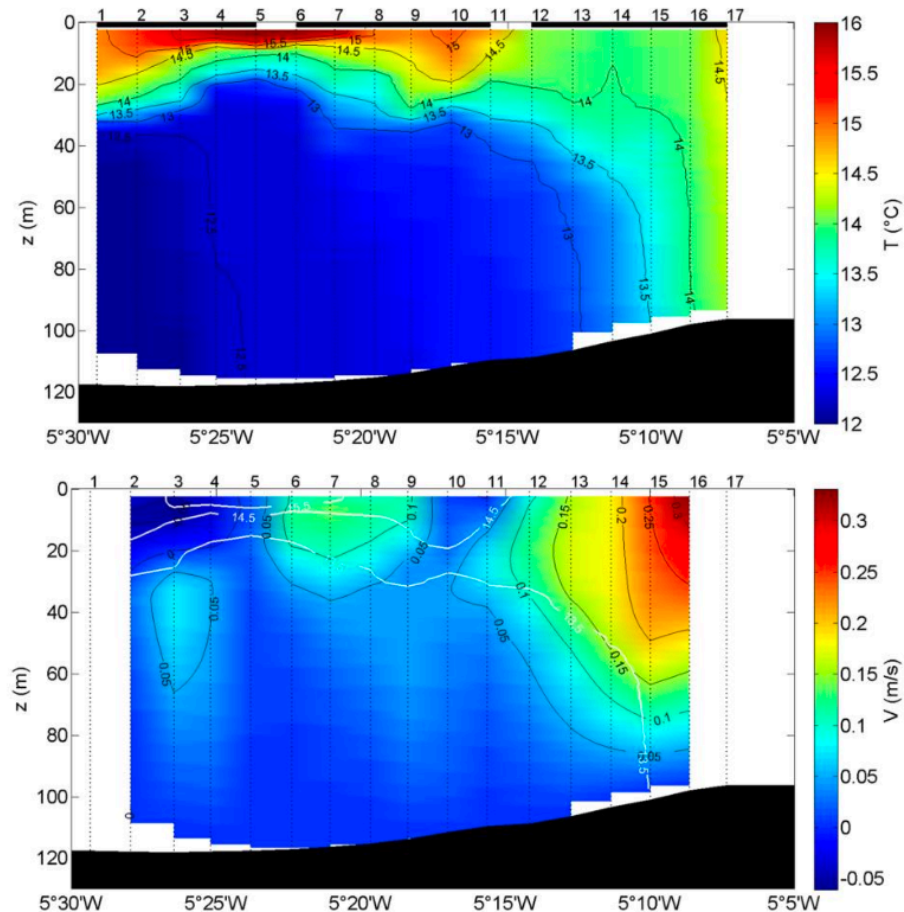
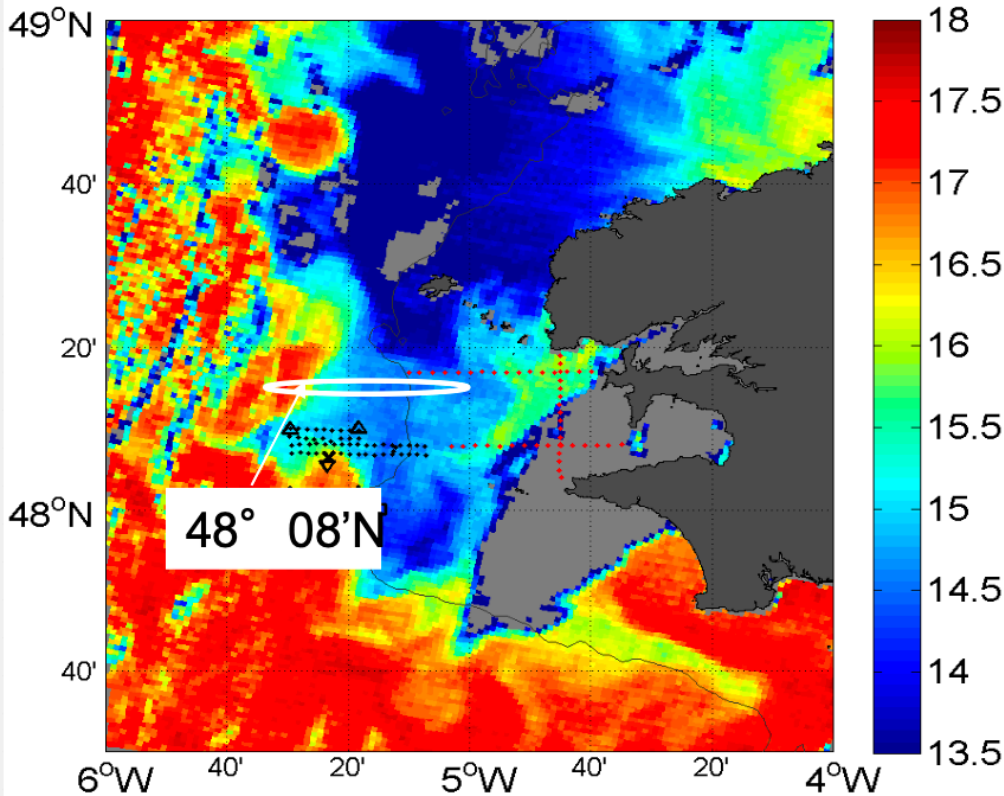
Dynamical aspects: Thermal wind

$$\partial_z v = -\frac{g}{\rho_0 f} \partial_x \rho$$

$$\partial_z u = \frac{g}{\rho_0 f} \partial_y \rho$$



thermal/tidal fronts



The interaction of the tidal wave with the shelf break give rise to an intensified mixed area where the stratification hardly sets on (due to this permanent mixing).

Geostrophic adjustment

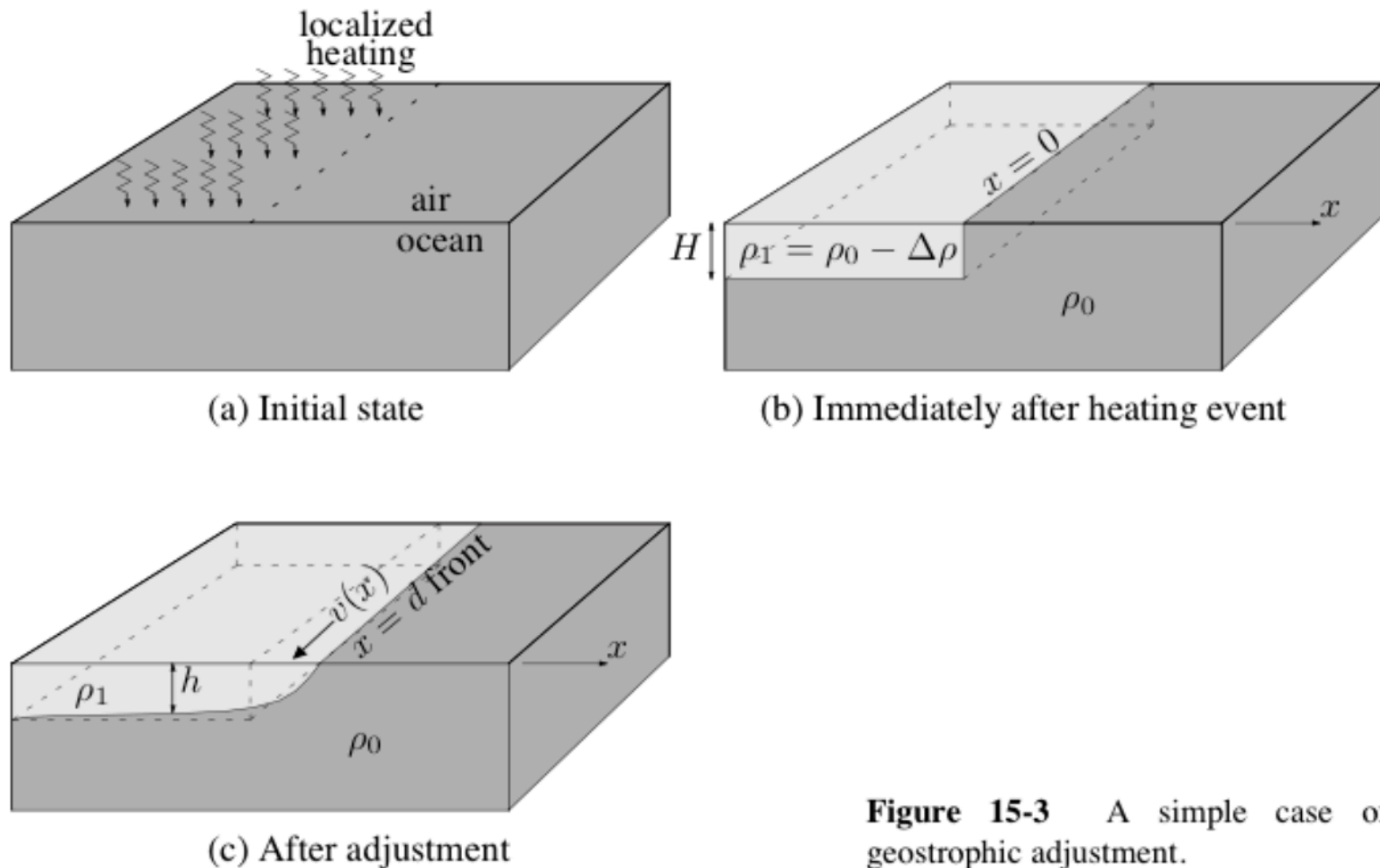
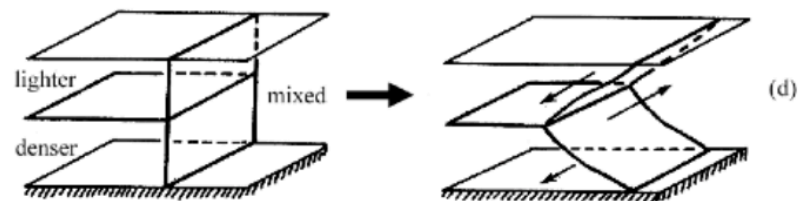
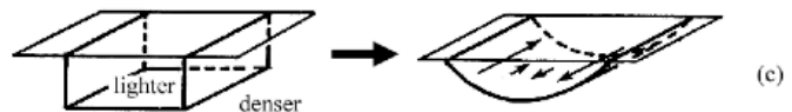
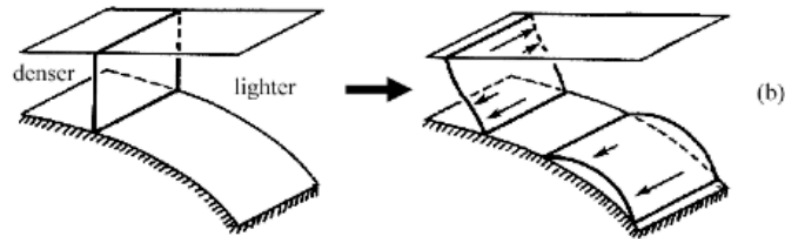
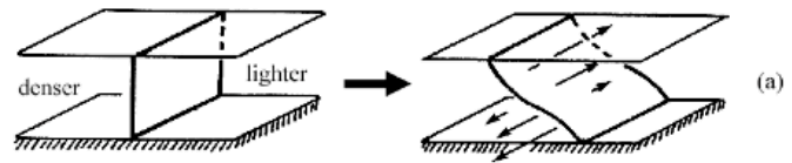


Figure 15-3 A simple case of geostrophic adjustment.

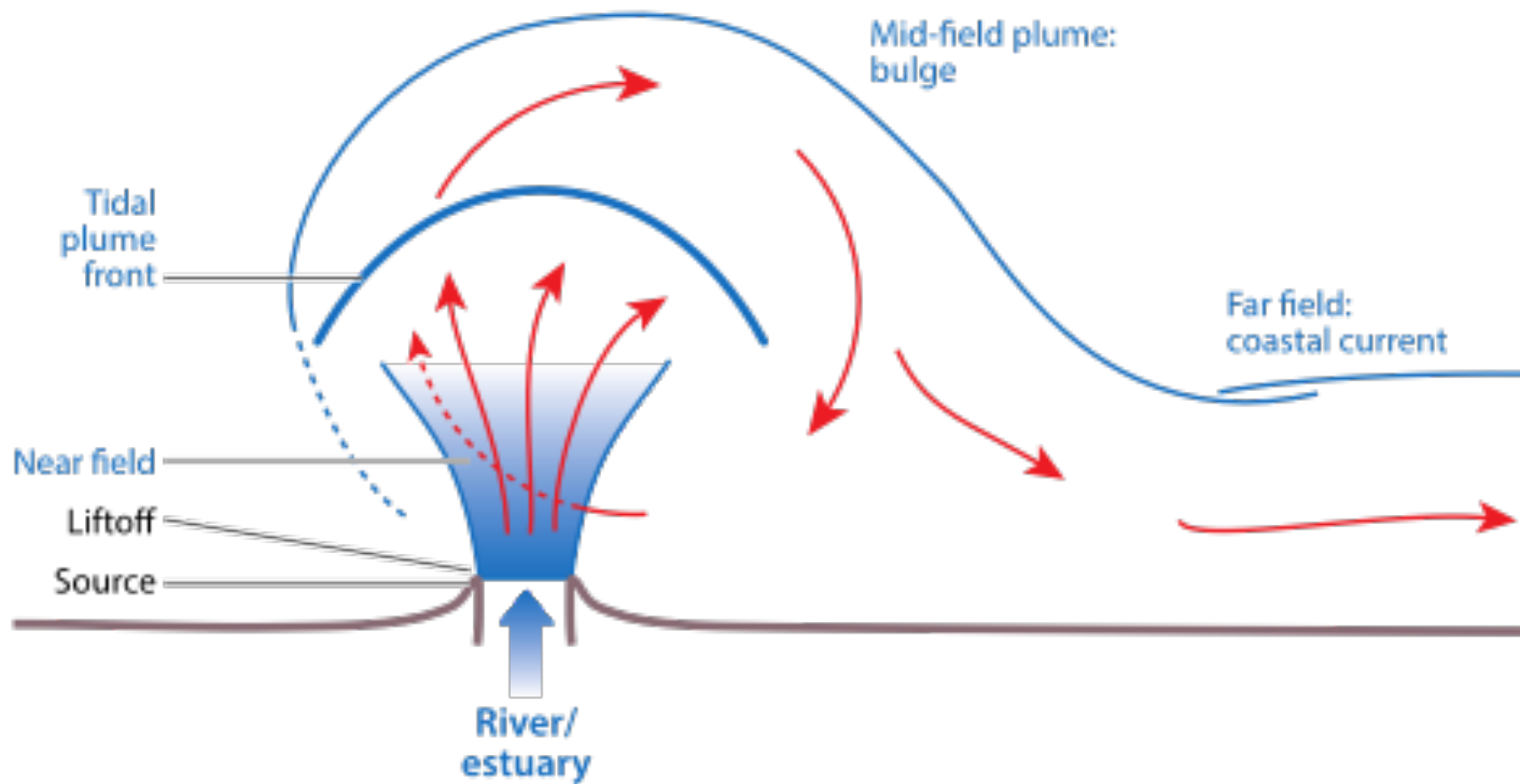
Geostrophic adjustment



River plumes

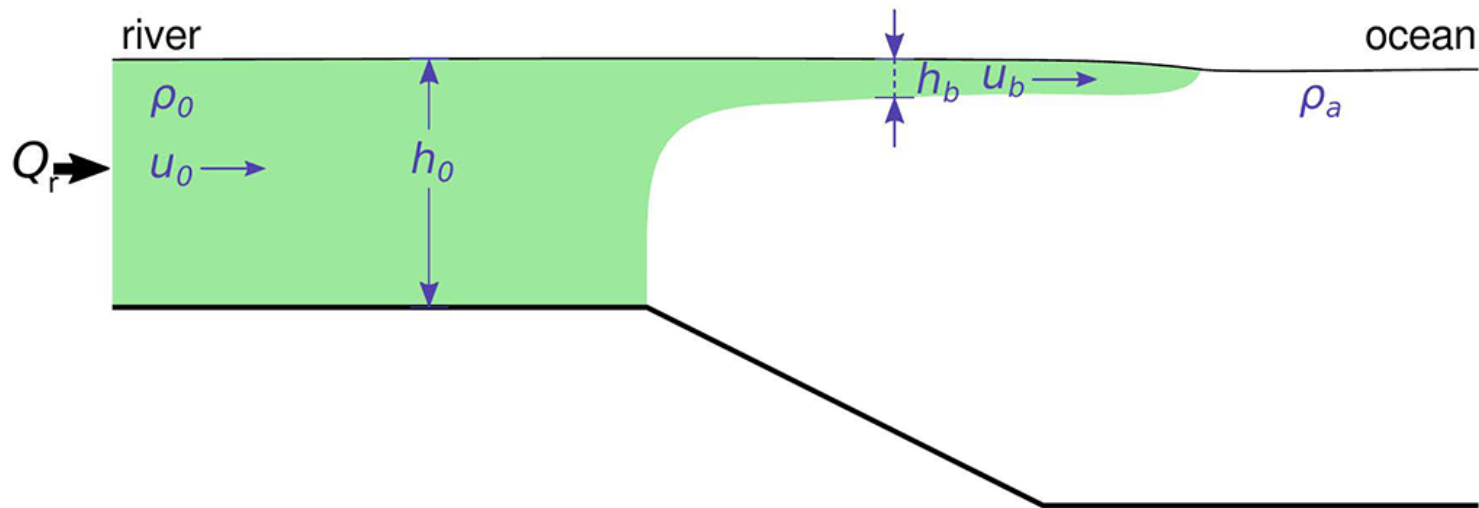


River plumes



Why does river water form a surface plume?

(b)



$$g' = g \frac{\rho_a - \rho_p}{\rho_0} > 0$$

ρ_a : ambient density ρ_p : plume density
 ρ_0 : reference density

Fresh water is usually less dense.
The density contrast drives spreading.

Which forces control the spreading?

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla_h) \mathbf{u} + f \hat{\mathbf{k}} \times \mathbf{u} = -g' \nabla_h h + \frac{\boldsymbol{\tau}}{\rho_0 h} + \mathbf{F}$$

Acceleration

Local change and advection of velocity.

Rotation

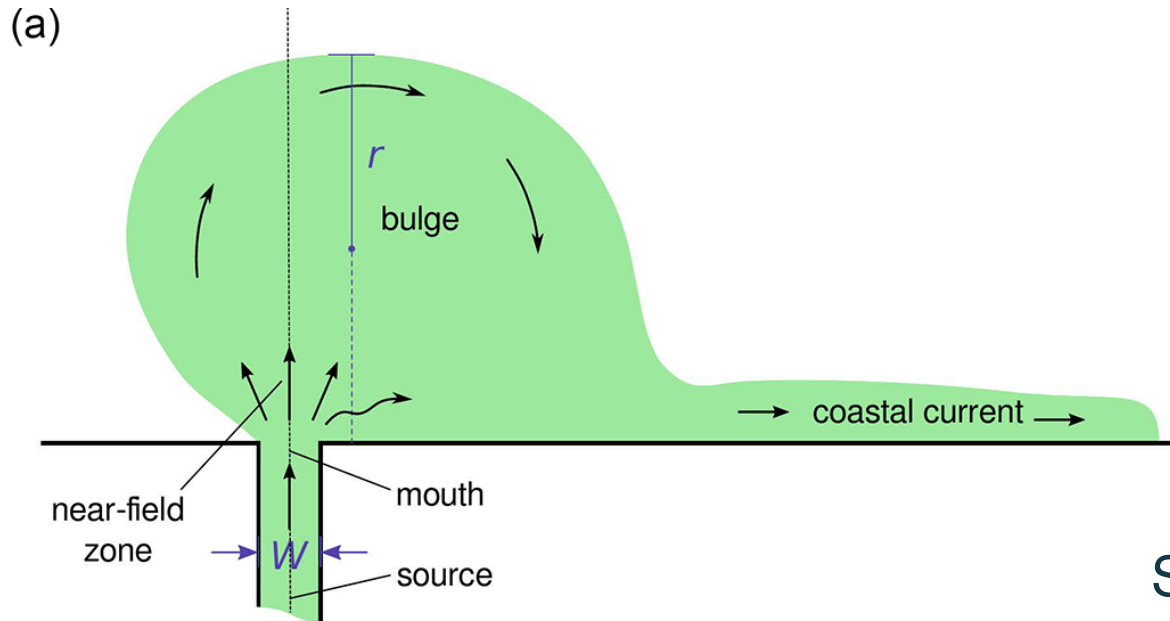
Coriolis redirects the flow.

Pressure + stress

Thickness gradients, wind and friction.

Thickness gradients drive flow from thicker to thinner regions.

Rotation turns spreading into a coastal current



$$f\hat{\mathbf{k}} \times \mathbf{u}_g = -g'\nabla_h h$$

$$Ro = \frac{U}{|f|L}, \quad R_d = \frac{\sqrt{g'h}}{|f|}$$

Small Ro : rotation matters.
 R_d : adjustment length scale.

A coastal current emerges as the plume adjusts to Earth's rotation.

Entrainment adds seawater and dilutes the plume

$$\frac{\partial h}{\partial t} + \nabla_h \cdot (h\mathbf{u}) = w_e$$

Volume enters the layer

$$\frac{\partial(hS_p)}{\partial t} + \nabla_h \cdot (h\mathbf{u}S_p) = w_e S_a$$

Ambient salt enters too

$$\frac{DS_p}{Dt} = \frac{w_e}{h} (S_a - S_p)$$

S_p rises toward S_a

$w_e > 0$: entrainment velocity S_p : plume salinity S_a : ambient salinity

Plume volume grows; freshwater is conserved

$$Q_f = Q_p \left(1 - \frac{S_p}{S_a} \right)$$

Freshwater transport = plume transport × freshwater fraction

$$S_a = 35, \quad S_p = 28 \quad \implies \quad Q_p = 5Q_f$$

The mixed plume carries 20% freshwater and 80% ambient seawater.

Q_f and Q_p : volume transports [$\text{m}^3 \text{s}^{-1}$] · Steady, uniform, one-way flow

Wind can spread or compress the plume

$$M_E = \frac{\tau \times \hat{\mathbf{k}}}{\rho_0 f}$$

$$U_E \sim \frac{|\tau|}{\rho_0 |f| h}$$

Upwelling-favourable wind

Offshore transport
Wider, thinner plume

Downwelling-favourable wind

Onshore transport
Narrower, deeper plume

For the same wind stress, a thinner plume responds more strongly.

τ : wind stress · M^E : transport [$\text{m}^2 \text{s}^{-1}$] · U^E : velocity scale [m s^{-1}]

River plumes

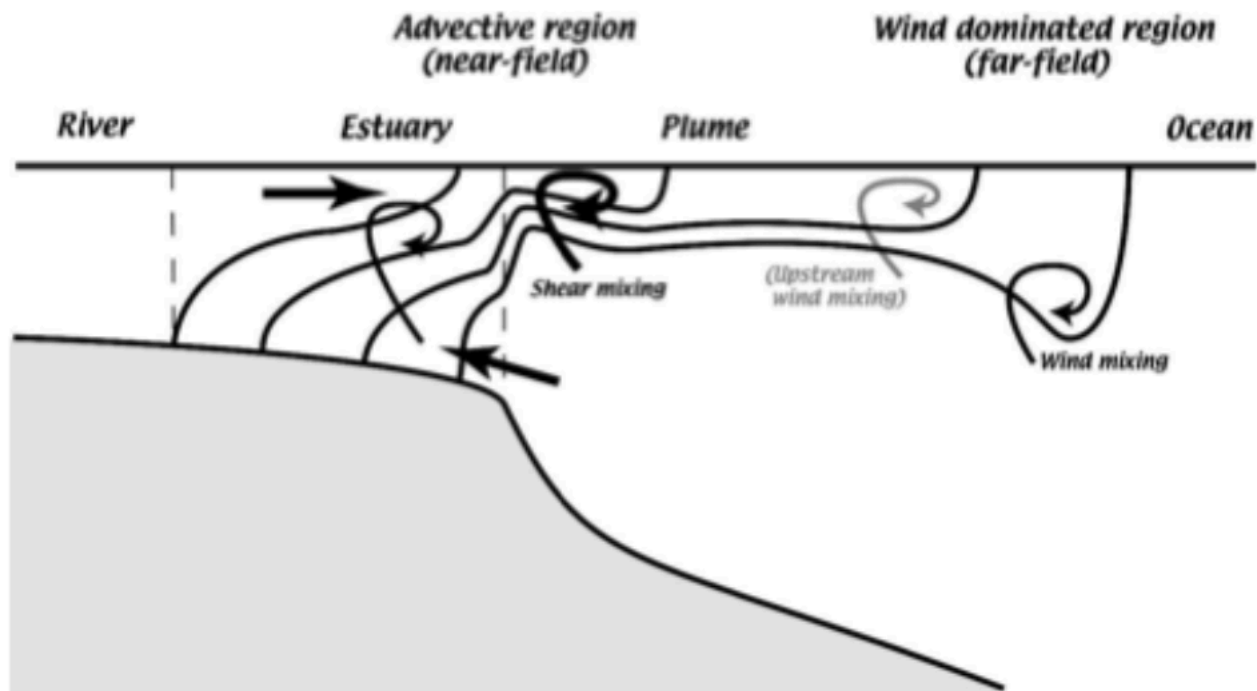
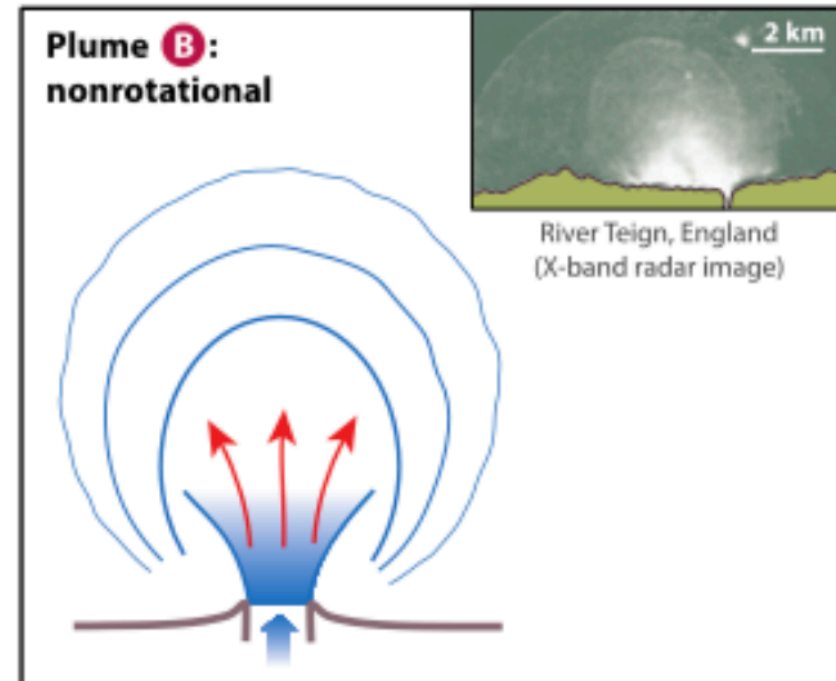
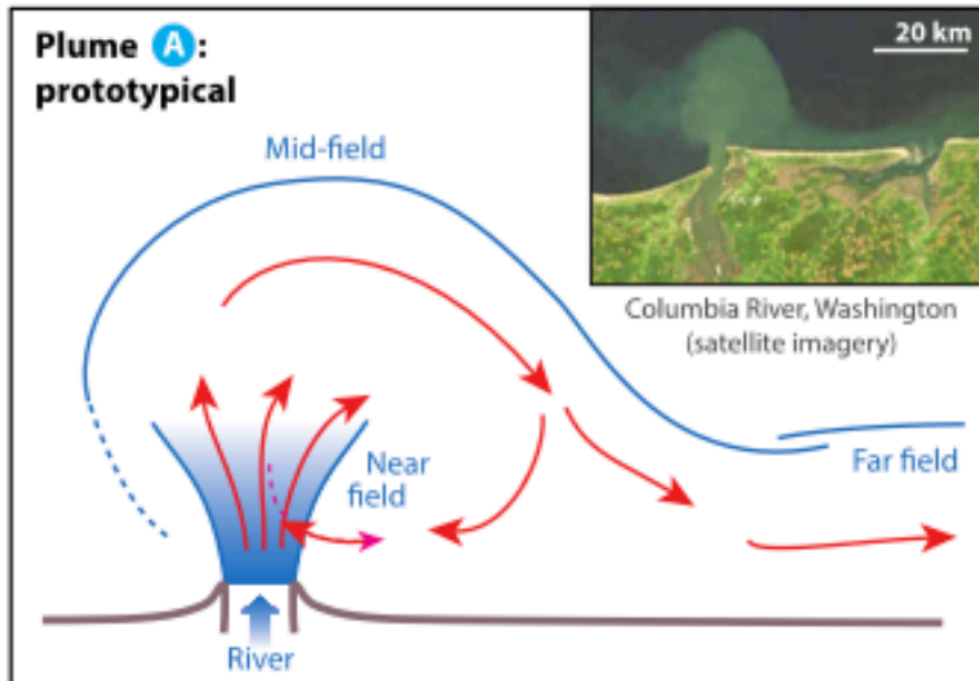


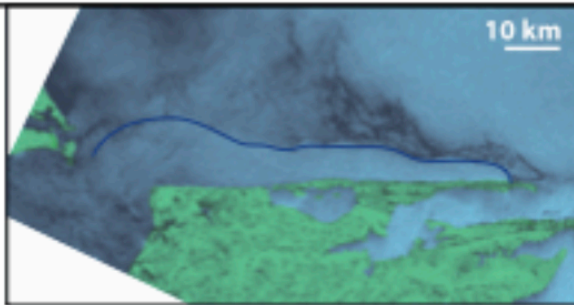
FIG. 1. This conceptual model of river plume anatomy shows the major regions and indicates the dominant mixing mechanisms.

A menagerie of plumes:

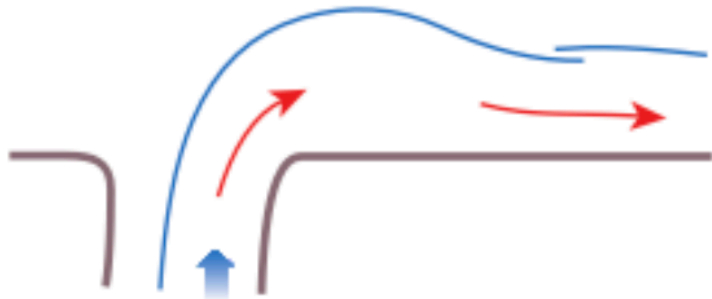


A menagerie of plumes:

Plume C:
wide estuary



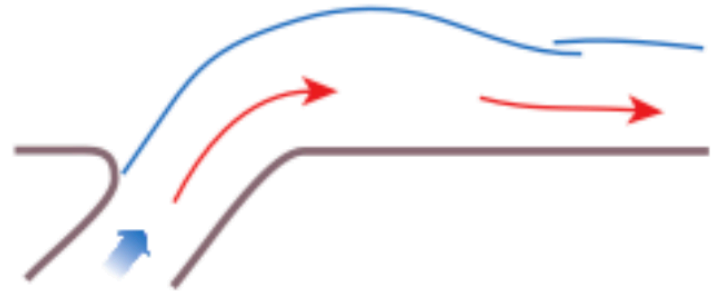
Chesapeake Bay, Virginia (SAR image)



Plume D:
angled inflow

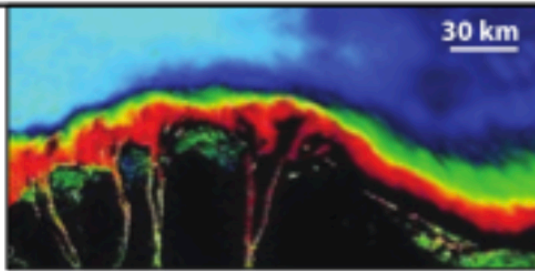


Eel River, California (rendering)

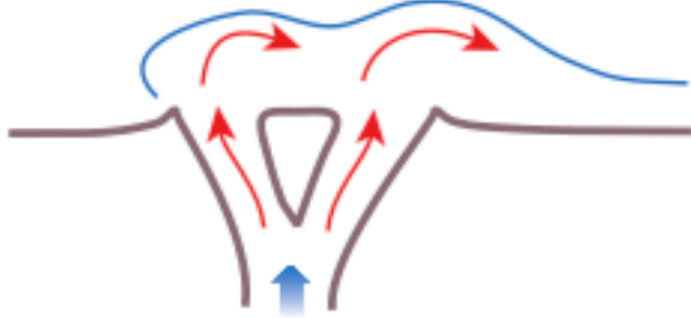


A menagerie of plumes:

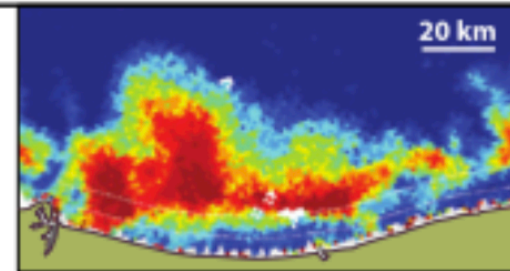
**Plume E:
delta**



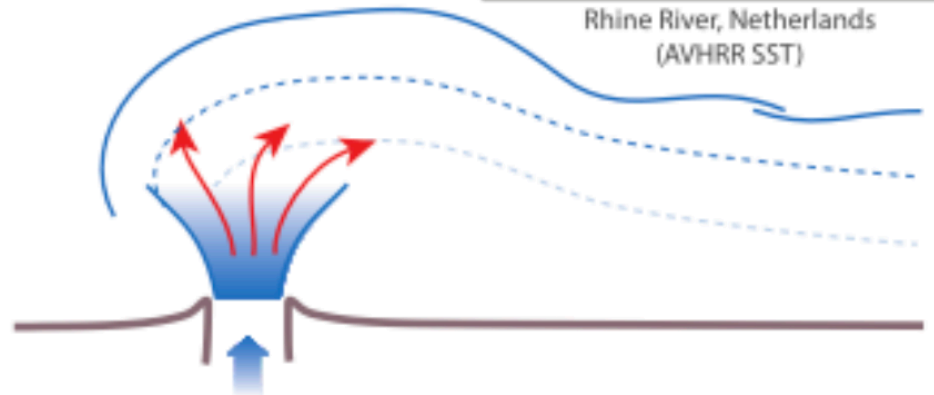
Mekong Delta, Vietnam
(satellite-derived turbidity)



**Plume F:
region of
freshwater
influence**



Rhine River, Netherlands
(AVHRR SST)



River plumes



Fraser River, 6 September 2014. NASA/JSC, Expedition 40.